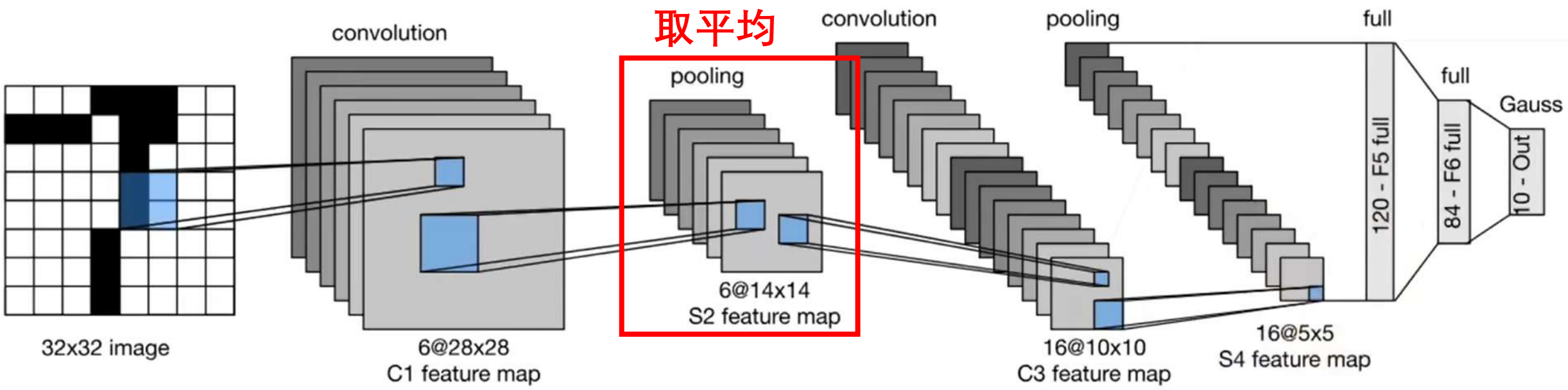
第6.3节 迁移学习

Category & Layer Transfer Learning

LeNet

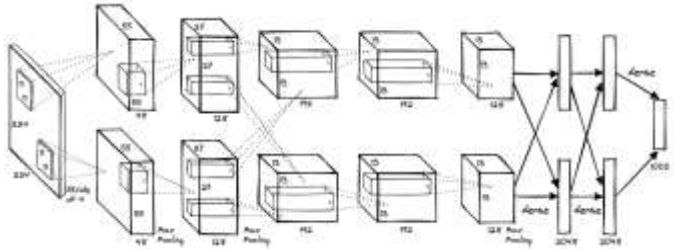


- LeNet是早期成功的神经网络
- 先使用卷积层来学习图片空间信息
- 然后使用全连接层来转换到类别空间

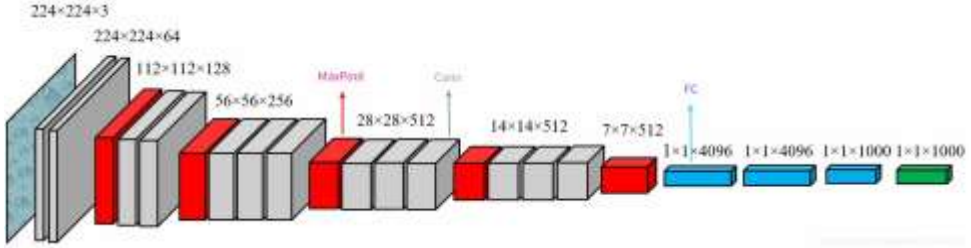
梯度消失使得深度神经网络难以训练

神经网络的能力和隐藏层数正相关

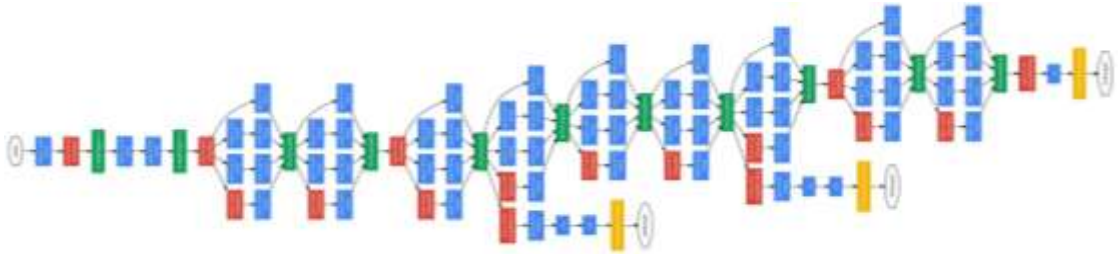
AlexNet'12
8 Layers
16.4%



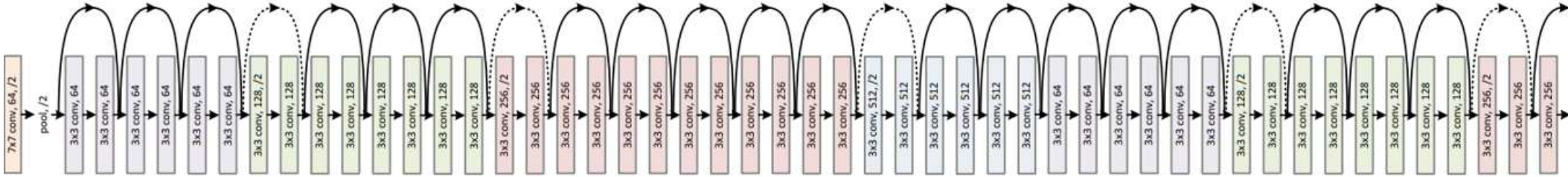
VGG'14
19 Layers
7.3%



GoogleNet'14
22 Layers
6.7%

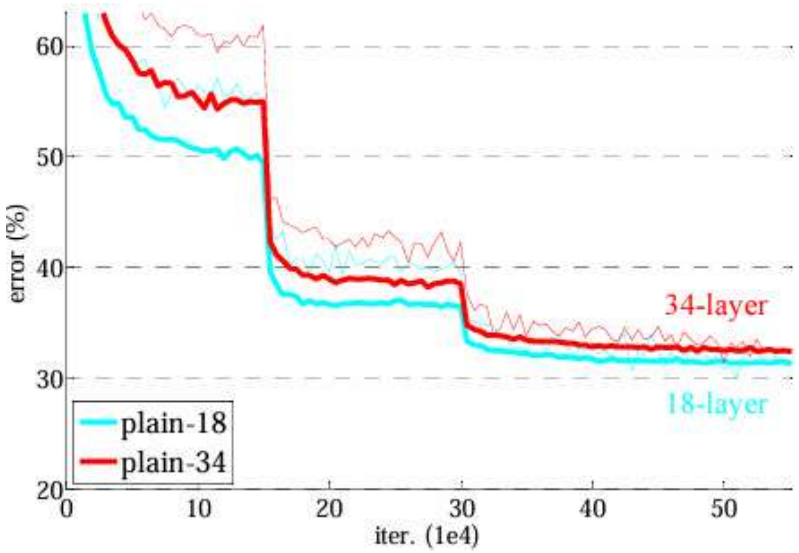


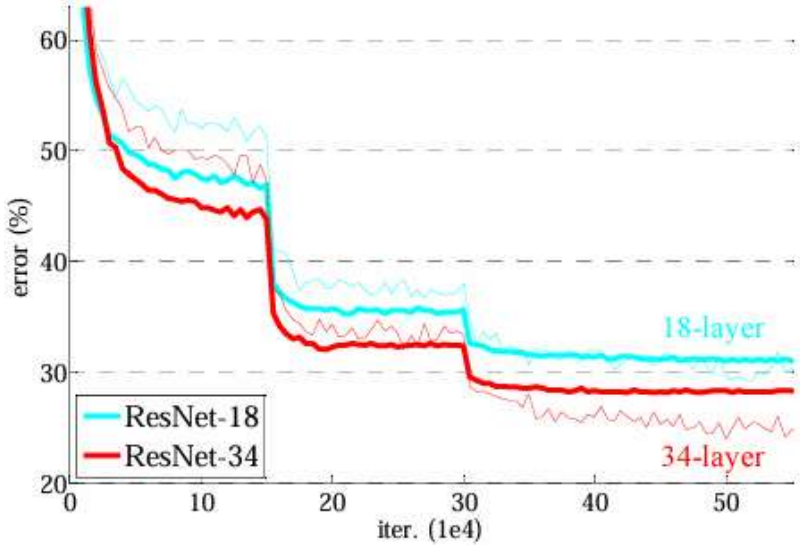
Residual Net'15
152 Layers
3.57%



ResNet让训练
真·深度神经网络
成为可能

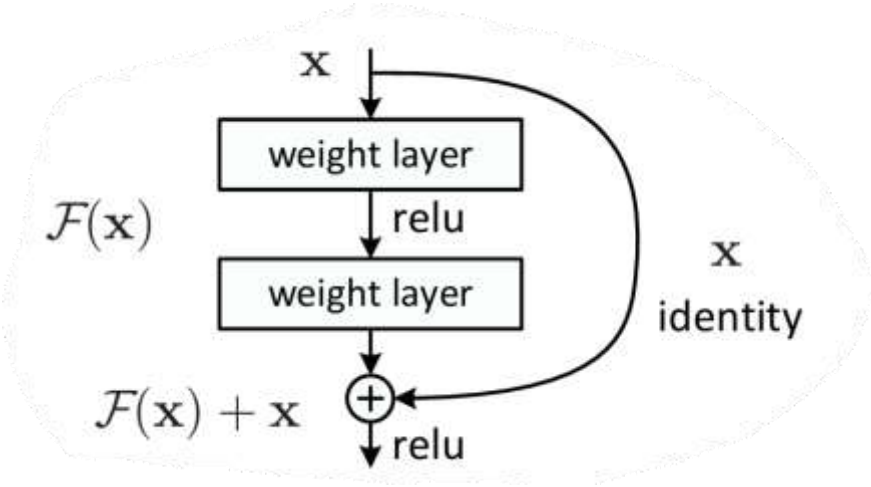
ResNet





ResNet朴素的愿望是

深模型至少不会比浅模型差



即使 F(x) 学“偏”了

x也不影响后续层的学习

IMAGENET （2010）

李飞飞

数据集	IMAGENET	MNIST
图片类型	自然物体的彩色图片	手写数字的黑白图片
图片大小	469 X 387	28 X 28
样本数量	1200000	60000
类数	1000	10

迁移学习

训练一个模型需要……

大量数据

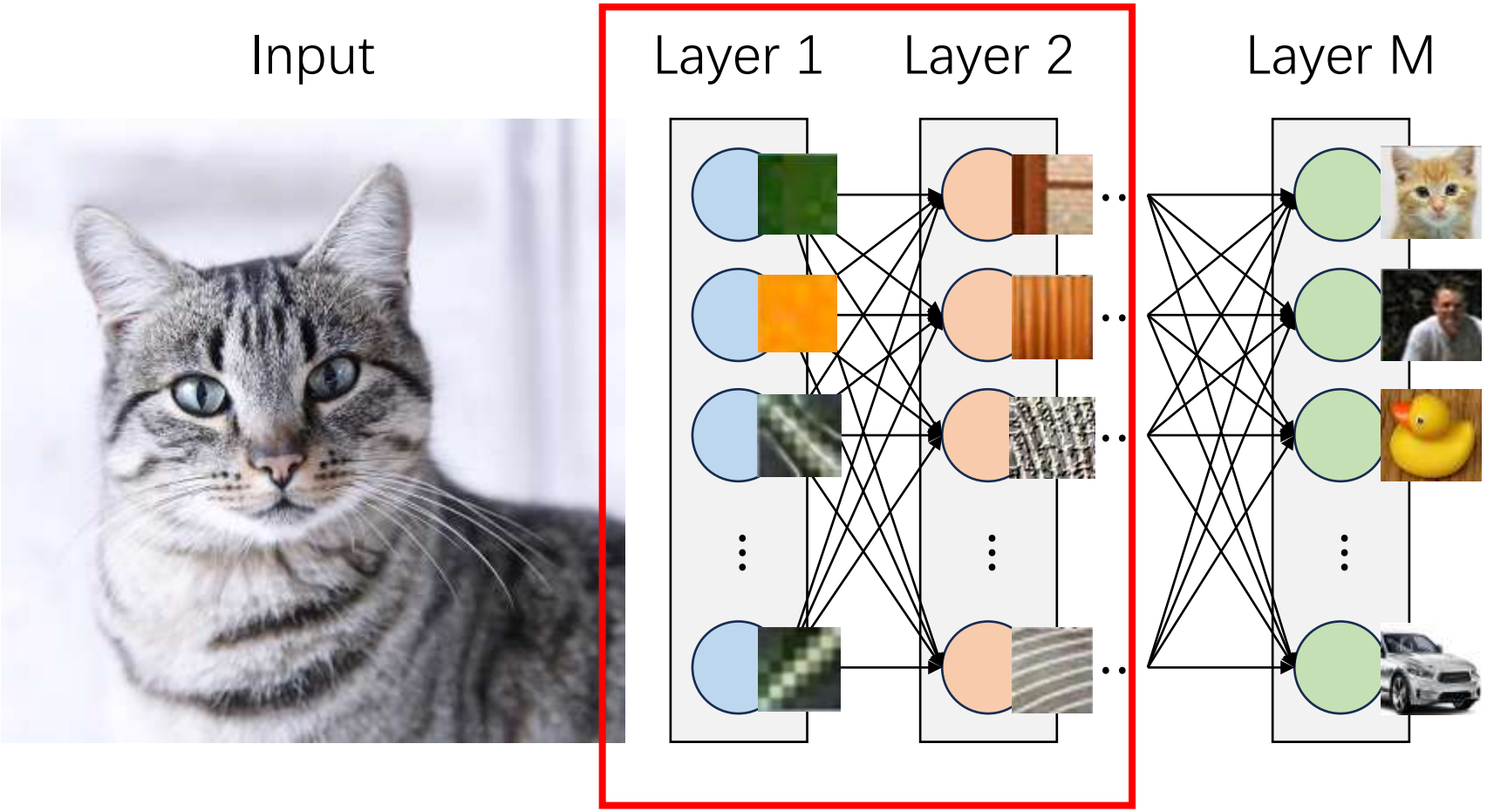
大量计算资源

然而实际上确只有……

小样本数据

有限计算资源

迁移学习



较低层的特征可能是不同目标所共有的

迁移学习

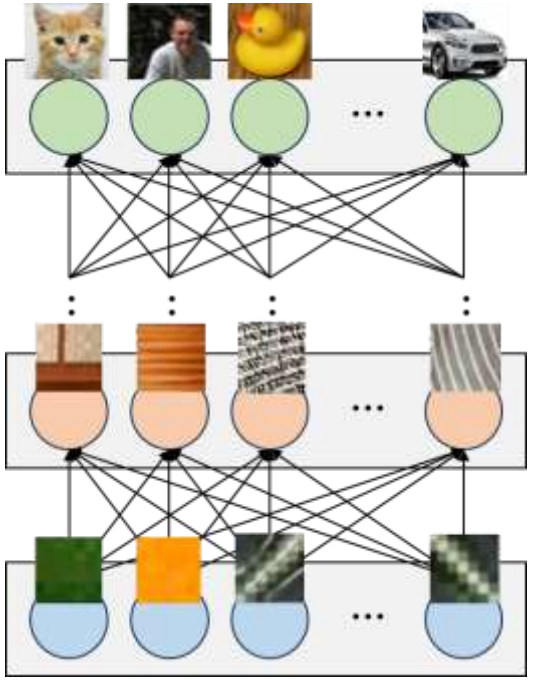
迁移学习的目的是迁移不同任务间共有的知识

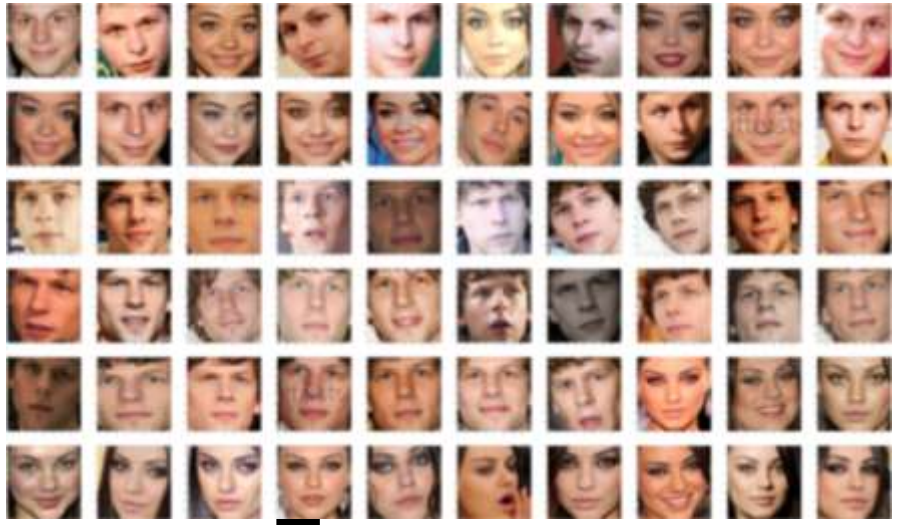
迁移底层

需学习的特征更小
对数据的要求更低

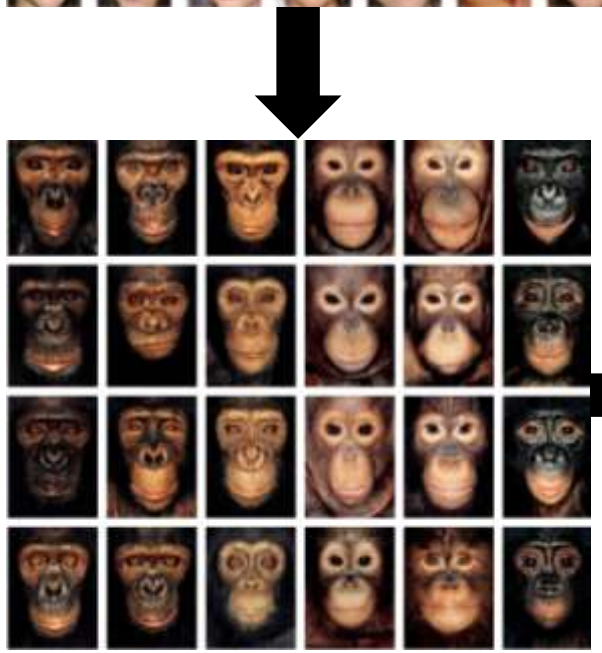
迁移学习

在大数据集上的预训练模型





迁移学习以少量的样本解决新任务

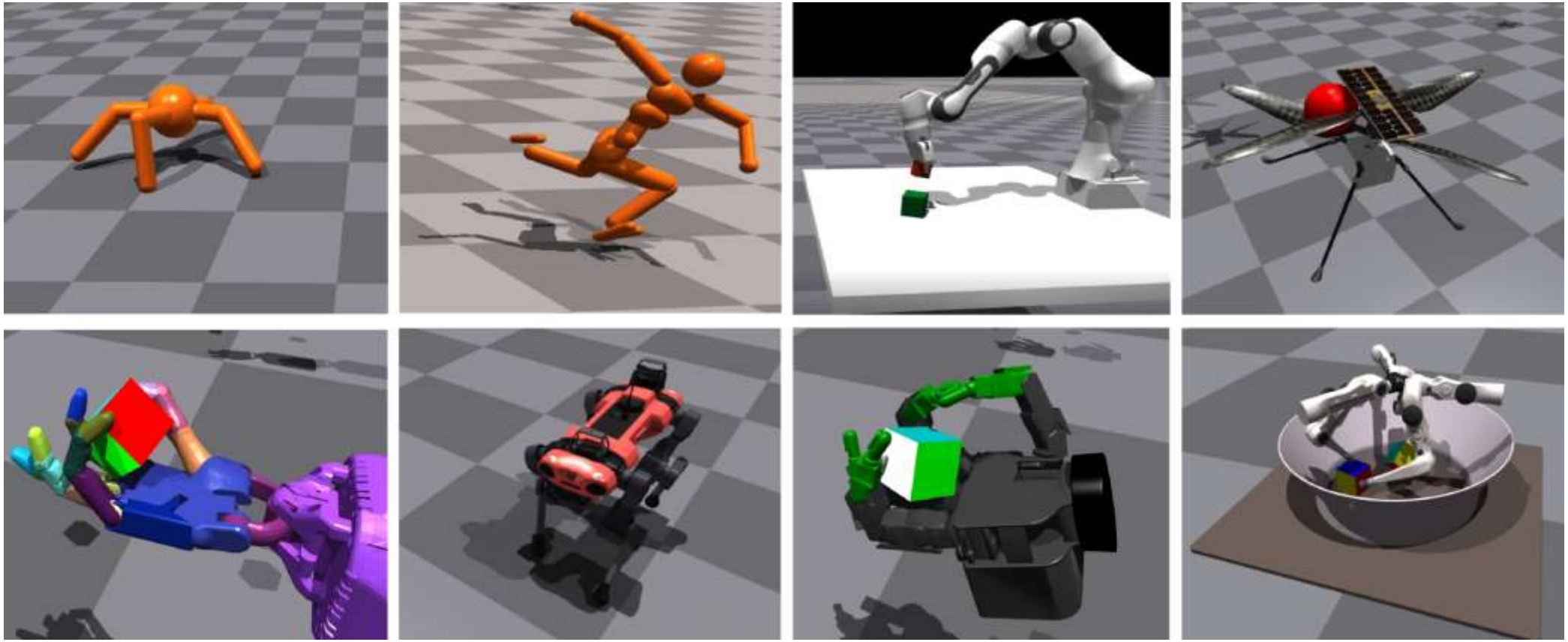


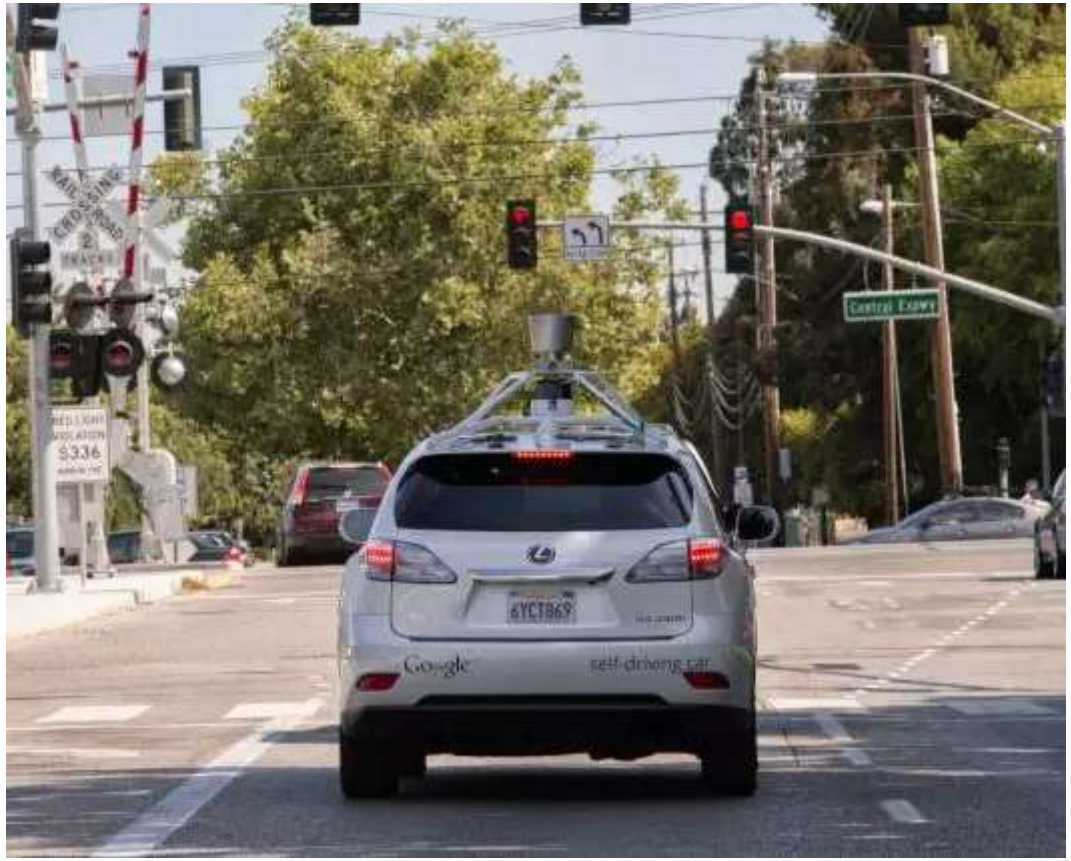
迁移学习

迁移不同语种之间的共性特征

迁移学习

迁移虚拟世界和物理世界的共性特征



迁移学习

迁移虚拟世界和物理世界的共性特征